Planning as a Cognitive Cost that Discourages Goal Switching

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Abstract

People exhibit a striking preference for maintaining their current goals even when better alternatives are readily available. While this bias can be framed as irrational (e.g., the sunk cost fallacy), goal persistence may be adaptive for conserving cognitive resources. We hypothesized that some of this persistence is attributable to the the cost of planning that comes with switching between goals. We conducted a behavioral experiment (n = 60) in which participants pursued goals under two conditions: one in which they had to plan towards goals before pursuing them and another in which they were given the optimal plan. We found that people persisted with their goals more when they had to create their own plans. We quantified the cognitive costs of planning with behavioral modeling as a function of decreased goal switching. These results indicate that people are sensitive to the act of planning, and suggest that goal persistence may be an adaptive response to these cognitive costs.

Introduction

People are notoriously unwilling to switch their goals. This “goal stickiness” can lead to suboptimal outcomes in situations when it would be more beneficial to switch to a similar but more feasible goal (Holton et al., 2024; Tomasello, 2022). While the sunk-cost fallacy looms as a prominent instance of irrational goal persistence (Arkes & Blumer, 1985; Ronayne et al., 2021), recent work has cast some goal perseveration as adaptive for managing cognitive costs (Holton et al., 2025). As goal-oriented computations (e.g., planning) are cognitively costly in terms of the time and effort involved in finding efficient solutions with respect to the given goal (e.g., the shortest route to a destination; Callaway et al., 2022; Kool et al., 2018), unwillingness to switch goals may therefore reflect rational conservation of these resources. We hypothesized that goal persistence may be, in part, driven by the costs of switching between goals and their associated plans.

Here, we test whether planning costs affect human goal abandonment and model goal switching using reinforcement learning (RL). Using a behavioral experiment, we show that people are less likely to abandon their goals when they have to form their own plans, as opposed to when already-formed plans are provided for them. Within-participant computational modeling captures this trend by treating planning costs as a value-based penalty. Overall, our results suggest that planning costs contribute to human goal persistence.

Methods

Behavioral Task

Participants (n = 60) were recruited via the online experimental platform Prolific. We tested participants’ selection and pursuit of goals using a "traveler’s task" (Figure 1) based on previous tasks such as the "fishing" and "suits" tasks (Aenugu & O’Doherty, 2024; Holton et al., 2024), where participants could make concurrent progress towards multiple tourist destinations (river, lake, and ocean). Travel to some destinations was easier than to others (there was more or less "traffic" on the way to each destination). Participants were notified when these traffic conditions changed periodically.

On each trial, participants were presented with a choice between 3 distinct goal destinations (Figure 1). Once participants chose their goal, they were presented with a 7×7 grid maze corresponding to their choice. Participants used their arrow keys to choose which direction (left/right/up/down) they would like to move. Based on a latent probability associated with the destination (i.e. "traffic"), they either moved one space in their chosen direction or remained in place. After displaying progress (or lack thereof), the trial ended, returning participants to the goal selection stage for the next trial. Over the course of many trials, participants could reach the destination. Once a destination was reached, the maze was reset to the start location and that destination could be reached again. Participants were instructed to try to reach as many destinations as possible.

We conducted a within-participant manipulation of the costs of planning between the two 90-trial blocks. In the "non-planning" condition, participants were shown and followed the shortest path route to the goal state. In the "planning" condition, participants had to develop their own optimal path to the goal location through planning.



Extended abstract presented at the *9th Conference on Cognitive Computational Neuroscience*, New York, NY, USA, 2026. Copyright 2026 by the author(s). Licensed under CC BY 4.0.

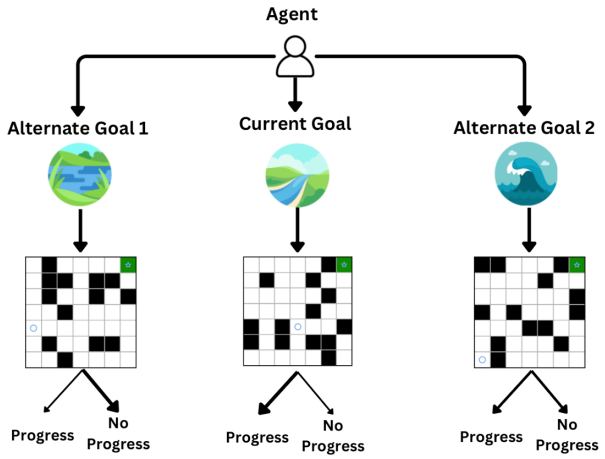


Figure 1: **Traveler Task.** Participants pursue completion of goals with varying progress rates.

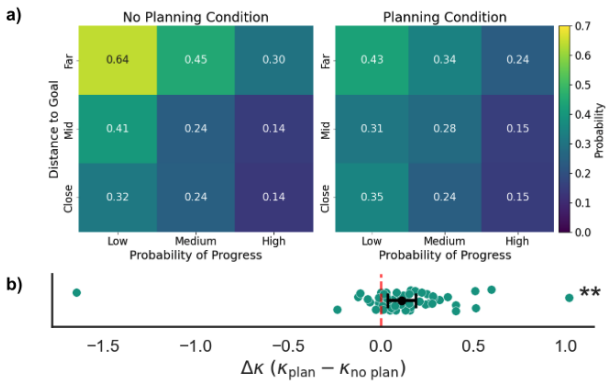


Figure 2: **a) Goal switch probabilities** for different underlying progress conditions, as a function of distance to the current goal; **b) Differences in fit κ values** between the planning and no planning condition, per participant. $\Delta\kappa > 0$ indicates a planning cost.

Modeling

People may persist on chosen goals because these goals have already yielded reward and thus accumulated learned value, or because they are more likely to have accumulated progress (Aenugu & O'Doherty, 2024; Holton et al., 2024). To account for these factors in isolating the cost of planning, we utilized the following goal valuation:

$$Q_t(g_i) = (\omega \cdot m_t(g_i) + (1-\omega))v_t(g_i) - c_t(g_i) \quad (1)$$

Here, $m_t \in [0,1]$ represents the proportion of progress made towards completion of goal g_i. The rate of goal progress $v_t(g_i)$ after being chosen at time t is learned using the Rescorla-Wagner learning rule $v_t(g_t) = v_{t-1}(g_t) + \alpha(r_t - v_{t-1}(g_t))$, where $\alpha \in (0,1)$ is the learning rate. The reward $r_t \in \{0,1\}$ indicates whether progress was made ($r_t = 1$) or not ($r_t = 0$). ω modulates the degree to which progress towards the goal so far influences the overall goal value $Q_t(g_i)$, with $\omega = 1$ indicating the most influence, while $\omega = 0$ indicates no influence. Finally, the choice kernel c_t acts as a switch cost capturing participant stickiness when switching between goals, where $c_t = 0$ when $g_i = g_{t-1}$ and $c_t = \kappa$ when $g_i \neq g_{t-1}$ and $\kappa \in [-1,2]$. Thus, $\kappa > 0$ quantifies goal stickiness beyond perseverance resulting from goal proximity and learned progress probability.

To fit the model to individual participant data, we used a softmax policy with inverse temperature $\beta \in [0,\infty)$ on learned goal Q values. The parameters β, α and two values for the choice kernel κ (κ_p for the planning condition and κ_{np} for the non-planning condition) were fit per participant. This model described the data better than a simple Rescorla-Wagner model (BIC = 316 worse than the current model) or a "fixed" momentum model with $\omega = 1$ (BIC = 702 worse).

Results

We found that people were significantly less likely to switch goals in the planning condition ($t = 3.17, p = 0.0023$). To visualize how goal switching behavior might depend on trial-by-trial probabilistic outcomes and distance made towards the goal, Figure 2a shows a 3×3 heatmap of the probability of goal abandonment based on a 3-part split of the distance to the goal and the *true* underlying probability of progress of the given goal (not the learned probability of progress) across participants, for each of the planning conditions. Figure 2a suggests that holding goal progress and progress probability constant, goal persistence increases when comparing the non-planning to the planning condition.

Using our model to isolate the costs of goal switching from the effects of learning and goal proximity on goal perseveration, we quantified the "cost" of planning (i.e. the component of stickiness that differs between conditions) as the difference between the two choice kernel parameters: $\Delta\kappa = \kappa_p - \kappa_{np}$. The mean $\Delta\kappa$ was significantly greater than 0 (Figure 2b, paired t-test, $t = 3.501, p < .001$).

Discussion

We tested whether the planning associated with switching goals contributes to human goal perseveration. We manipulated the ease of planning and controlled for learning and goal-proximity effects which suggested that the act of planning contributes to this perseveration. While this supports a growing body of research regarding how goal commitment support efficient use of cognitive resources (Callaway et al., 2022; Holton et al., 2024; Kool et al., 2018), future work can investigate what aspects of forming plans lead to goal perseveration (e.g., cost avoidance vs. preferring self-constructed plans).

Acknowledgments

This research was supported by Princeton University's Office of Undergraduate Research Student Initiated Internship Program (OURSIP) through the Hewlett Foundation. Generative AI (Microsoft Copilot) was used to review code for errors.

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